

Computer Programming with MATLAB



Lesson 2: Matrices and Operators

by

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Introduction to Arrays and Matrices

▶ Array

- Any set of numbers arranged in a rectangular pattern.

Example—

A page with six rows of four numbers each is a two-dimensional array

10	14	48	25
24	34	17	35
22	33	29	44
32	8	11	48
35	6	37	27
37	25	13	7

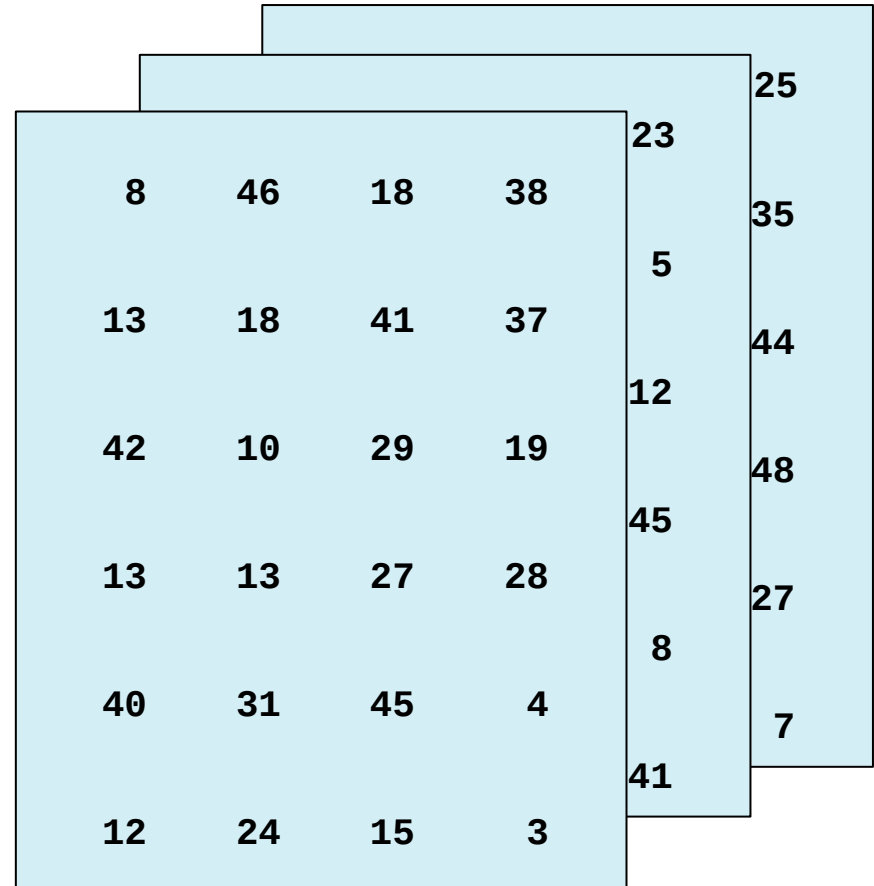
Introduction to Arrays and Matrices

▶ Array

- Any set of numbers arranged in a rectangular pattern.

Three-dimensional
Example—

A stack of such
pages



8	46	18	38	23	25
13	18	41	37	5	35
42	10	29	19	12	44
13	13	27	28	45	48
40	31	45	4	8	27
12	24	15	3	41	7

Introduction to Arrays and Matrices

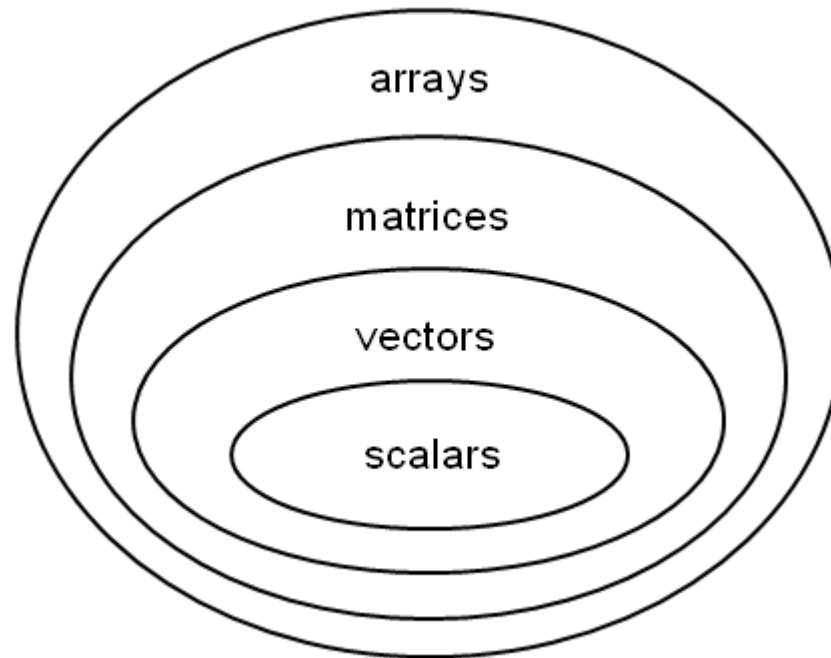


- ▶ Higher dimensions are uncommon
- ▶ The most common have special names:
 - 2D array = “matrix” (plural is “matrices”)
 - 1D array = “vector”
- ▶ Most ingenious part of Cleve Moler’s invention of MATLAB was the way he set it up to deal with matrices.
- ▶ MATLAB stands for “Matrix Laboratory”!

Arrays and Matrices



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Rows and Columns

```
>> X = [1:4; 5:8; 9:12];
```

1:	1	2	3	4
2:	5	6	7	8
3:	9	10	11	12

rows

Rows and Columns

```
>> X = [1:4; 5:8; 9:12];
```

1:	2:	3:	4:
1	2	3	4
5	6	7	8
9	10	11	12

columns



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Indexing

```
>> X = [1:4; 5:8; 9:12];
```

```
>> X(2,3)
```

3:

	1	2	3	4
2:	5	6	7	8
	9	10	11	12

```
>> ans =
```

7



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Array Addition

- ▶ $Z = X + Y$ means
 - $Z(m,n) = X(m,n) + Y(m,n)$ for all valid m and n

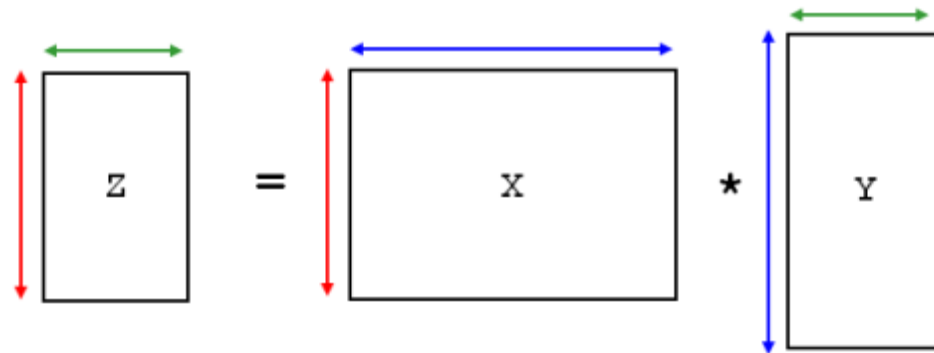
$$\begin{array}{lcl} Z(1, 1) & = & X(1, 1) + Y(1, 1) \\ Z(1, 2) & = & X(1, 2) + Y(1, 2) \\ \dots & & \\ Z(1, \text{end}) & = & X(1, \text{end}) + Y(1, \text{end}) \end{array} \left. \vphantom{\begin{array}{l} Z(1, 1) \\ Z(1, 2) \\ \dots \\ Z(1, \text{end}) \end{array}} \right\} \text{1st row}$$
$$\begin{array}{lcl} Z(2, 1) & = & X(2, 1) + Y(2, 1) \\ Z(2, 2) & = & X(2, 2) + Y(2, 2) \\ \dots & & \\ Z(2, \text{end}) & = & X(2, \text{end}) + Y(2, \text{end}) \end{array} \left. \vphantom{\begin{array}{l} Z(2, 1) \\ Z(2, 2) \\ \dots \\ Z(2, \text{end}) \end{array}} \right\} \text{2nd row}$$
$$\dots$$
$$\begin{array}{lcl} Z(\text{end}, 1) & = & X(\text{end}, 1) + Y(\text{end}, 1) \\ Z(\text{end}, 2) & = & X(\text{end}, 2) + Y(\text{end}, 2) \\ \dots & & \\ Z(\text{end}, \text{end}) & = & X(\text{end}, \text{end}) + Y(\text{end}, \text{end}) \end{array} \left. \vphantom{\begin{array}{l} Z(\text{end}, 1) \\ Z(\text{end}, 2) \\ \dots \\ Z(\text{end}, \text{end}) \end{array}} \right\} \text{last row}$$

Matrix Multiplication

- ▶ Different from Array Multiplication!
- ▶ $Z = X * Y$ means that for all valid m and n

$$Z(m, n) = \sum_k X(m, k) Y(k, n)$$

- ▶ Not always legal:
 - Inner dimensions of X and Y must be the same



Array Division

- ▶ **$Z = X ./ Y$**
 - means that for each m and n , $Z(m,n) = X(m,n)/Y(m,n)$
- ▶ **$Z = X .\ Y$**
 - means that for each m and n , $Z(m,n) = Y(m,n)/X(m,n)$
- ▶ Try these out in MATLAB on your own!
- ▶ Matrix division is a complicated concept in linear algebra, so we are not covering it here
 - But you can check out the advanced concepts of the textbook for detailed explanation

Precedence

- ▶ $x = a + b + c$
 - order does not matter with addition
- ▶ $y = c + a * b$ is not the same as
- ▶ $y = (c + a) * b$
- ▶ Multiplication has priority over addition
 - In programming, this is called *precedence*

PRECEDENCE	OPERATOR
0	Parentheses: (...)
1	Exponentiation ^ and Transpose '
2	Unary +, Unary -, and logical negation: ~
3	Multiplication and Division (array and matrix)
4	Addition and Subtraction
5	Colon operator :

Precedence Table

Associativity

- ▶ $x = a + b + c$
- ▶ $x = a * b * c$
 - order does not matter with addition or multiplication
- ▶ $y = a \wedge (b \wedge c)$ is not the same as
- ▶ $y = (a \wedge b) \wedge c$
- ▶ In programming, the order in which operators of the same precedence are executed is called *associativity*
- ▶ In MATLAB, it is left to right
- ▶ $y = a \wedge b \wedge c$ is the same as
- ▶ $y = (a \wedge b) \wedge c$